



JOMON GEORGE REJI

Mechanical & Mechatronics Engineer

Nuremberg, Germany | Date of Birth: 03.02.1999, India

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Portfolio | GitHub | LinkedIn

Professional Profile

Master's student in **Mechatronics and Cyber-Physical Systems** at **Technische Hochschule Deggendorf** with a Bachelor's degree in **Mechanical Engineering**. Strong focus on **Autonomous Systems, ROS2-based Robotics, SLAM, Sensor Perception, and Control System Design**.

Hands-on experience designing and implementing complete **ROS2 autonomous robotic systems**, including SLAM-based mapping, sensor fusion, path planning, and simulation-based validation. Proficient in **Python and C++**, confident with Linux and Git, with additional experience in MATLAB/Simulink-based control system design. Analytical and structured working style with strong technical documentation skills, developed through coursework, independent projects, and industrial engineering experience.

Education

M.Eng. Mechatronics and Cyber-Physical Systems

March 2023 – Present

Technische Hochschule Deggendorf, Germany

Relevant Coursework: Autonomous Systems, Robotics, Sensor Systems, Artificial Intelligence, Machine Learning, Control Systems, Simulation Technologies

B.Tech. Mechanical Engineering

August 2017 – June 2021

APJ Abdul Kalam Technological University, India

CGPA: **8.09/10**

Relevant Coursework: Machine Design, Finite Element Analysis, CAD/CAM, Manufacturing Engineering, Thermal Engineering, Strength of Materials

Professional Experience

Junior Engineer Trainee – Sharda Motors Pvt. Ltd., India

November 2022 – January 2023

- Supported manufacturing, testing, validation, and quality assurance in an industrial environment.
- Assisted in engineering documentation and process optimisation.

Graduate Mechanical Trainee – Newfriday Engineering Pvt. Ltd., India

January 2022 – November 2022

- Supported engineering design, technical documentation, and production workflows.
- Manufacturing and assembling of mechanical parts and spares.

Technical Skills

Programming: Python, C++, MATLAB

Robotics & Navigation: ROS, ROS2, SLAM, HD Mapping Concepts, LiDAR-Based Mapping, Localisation, Trajectory Estimation, Autonomous Navigation (Nav2), Path Planning, Mobile Robotics, Robot Perception

Sensor Perception & Fusion: LiDAR, Sensor Fusion, Camera-LiDAR Fusion, Multi-Sensor Data Synchronisation, Sensor Calibration & Filtering, State Estimation, Probabilistic Robotics (Academic)

Simulation & Evaluation: Gazebo, MATLAB/Simulink, Unity, Digital Twin Concepts, Performance Metrics (Trajectory Error, Robustness Evaluation)

Software Development: Object-Oriented Programming (OOP), Algorithm Design, Git, Linux, Debugging, Technical Documentation, Modular ROS2 Node Architecture

Engineering Tools: Arduino, Visual Studio Code, CAD, FEA

Relevant Projects

Autonomous Robot – SLAM, Mapping & Multi-Sensor Perception (ROS2)

- Designed and implemented a complete **SLAM and mapping pipeline** in ROS2 using LiDAR data: localisation, environment mapping, and autonomous navigation.

- Implemented **sensor fusion** combining LiDAR with additional sensor inputs for robust environment perception under varying conditions.
- Performed data synchronisation, filtering, and calibration of sensor streams to ensure consistent mapping accuracy; evaluated results using trajectory error and map quality metrics.
- Designed modular **ROS2 software architecture** in Python and C++, maintained source code with Git, and produced structured technical documentation of methods and results.
- Validated system robustness across varying environment scenarios in Gazebo simulation, including narrow corridors and dynamic obstacles.

Dual-Axis Solar Tracking System (MATLAB/Simulink)

- Designed a sensor-driven **closed-loop control and state estimation system** with real-time tracking and simulation-based validation and optimisation.
- Modelled dynamic actuator behaviour and tuned controller parameters to minimise tracking error while maintaining stability under disturbance conditions.

VR Industrial Simulation Environment (Unity)

- Developed 3D simulation environments for industrial training scenarios, demonstrating spatial modelling and scenario-based evaluation skills.
- Designed interactive scenarios around common safety procedures and failure modes, balancing realism with usability.

Safety Click Razor – Sustainable Design (Additive Manufacturing)

- Designed and optimised a sustainable product using CAD and FEA-based structural validation; structured engineering documentation methodology.
- Iterated component design based on FEA results to balance material efficiency against mechanical reliability under additive-manufacturing-specific structural behaviour.

Languages

English (Fluent, C1) German (A2 – actively pursuing B1) Malayalam (Native)

Certifications

Introduction to Programming with MATLAB – Coursera, 2022
AI For Everyone – Coursera, 2022
Aerospace Materials – Coursera, 2022
Product Design Using CATIA V5 – CADD Centre Training Services Pvt Ltd., 2019
Mechanical Engineering – Internal Combustion Engine Basics – Alison, 2019

Research Interests

SLAM & Mapping ● Multi-Sensor Perception ● Autonomous Navigation ● Trajectory Estimation & Evaluation ● Probabilistic Robotics ● ROS2 System Design ● Control Systems ● Industrial Automation